Topic 36 — Ozone Layer

Covers syllabus (click to expand)

Ozone Layer | Ozone Depletion (ओजोन क्षरण) | Ozone Hole (ओजोन छिद्र) | Ozone Depleting Substances (ODS) | Montreal Protocol (मॉन्ट्रियल प्रोटोकॉल) | Vienna Convention

Sources baked in: NCERT; UNEP Ozone Secretariat; Vienna 1985; Montreal 1987 (force 1989); Kigali 2016; MoEFCC Ozone Cell; UPPCS/UPSC PYQs **Weight:** ★★★ High — Montreal ≠ Kyoto; ozone = stratosphere; Ozone Day 16 Sept; CFC link **Last verified:** September 2026

Current Affairs (this topic)

- Twin of Ozone Layer core — keep Montreal / Kigali as ozone-family CA only.
- Prefer one consistent ODS story across both ozone chapters.

36.1 Ozone Layer

Identity: Stratosphere ~15–30 km | UV shield | Dobson Unit | Chapman cycle | NO_2 regulation

- **Ozone (O_3)** is triatomic oxygen that absorbs harmful solar ultraviolet radiation.
- The protective **ozone layer** sits mainly in the lower **stratosphere** (~**15–30 km**, sometimes cited to ~**35 km**; peak near ~**25 km**). The stratosphere itself spans roughly **10–50 km**.
- About **90%** of atmospheric ozone is stratospheric; about **10%** is tropospheric.
- Natural formation/destruction follows the **Chapman cycle**: UV splits $O_2 \rightarrow O$; $O + O_2 \rightarrow O_3$; UV also breaks O_3.
- Stratospheric ozone is naturally regulated in part by **nitrogen dioxide (NO_2)**.
- Column thickness varies seasonally — often **highest in spring** and **lowest in the rainy season** (general pattern; the Antarctic “hole” is a separate spring-extreme story).
- **UV-A (315–400 nm)** largely reaches the surface; **UV-B (280–315 nm)** is mostly blocked by ozone; **UV-C (100–280 nm)** is blocked by O_2/O_3 and does not reach the surface under normal conditions.
- Stratospheric ozone blocks roughly **93–99%** of the most damaging solar UV.
- Near the surface, ozone is a **poisonous pollutant** (lungs/respiratory harm). In the stratosphere it is essential because it shields life from UV.
- Ozone absorption heats the stratosphere, so temperature **rises with height** there.
- Column ozone is measured in **Dobson Units (DU)**; **1 DU ≈ 0.01 mm** of pure ozone at **0°C and 1 atm**. Global average ≈ **300 DU**. Total atmospheric ozone mass is on the order of **~3 billion tonnes**.

PYQ — UPPCS Prelims 2025, Q45

1. Ozone layer protects Earth’s surface from solar UV.

2. Ozone depletion has been linked to CFCs.

Options: A. Only 2 B. Neither C. Both 1 and 2 D. Only 1

Show answer

Ans: C

PYQ — UPPCS Prelims 2018, Q52 / Q57

Maximum ozone concentration / wrong pair:

Show answer

Stratosphere. Pair **Ozone layer — Troposphere** is NOT correctly matched.

PYQ — UPPCS Prelims 2018, Q55

Ultra-violet radiation from sunlight causes the reaction that produces:

Options: A. CO B. SO_2 C. O_3 D. Fluorides

Show answer

Ans: C

> **Logic:** UV-shield ozone = **stratosphere**. Ground O_3 = bad. Ozone absorbs **UV**, not IR as its protective role.

36.2 Ozone Depletion

Identity: ODS → Cl·/Br· | Catalytic destruction | Rowland–Molina 1974 | CFC uses

- **Ozone depletion** is a sustained fall in stratospheric ozone below natural levels.
- Stable **ODS** (especially CFCs) reach the stratosphere, where UV releases **chlorine** (or bromine) radicals.
- One chlorine atom can destroy on the order of **~100,000** ozone molecules in a catalytic cycle.
- **Rowland–Molina (1974)** linked CFCs to ozone loss; Nobel Prize in Chemistry **1995**.
- CFCs are non-toxic, non-flammable, long-lived (~**80–100 years**) coolants/solvents — also strong **greenhouse gases**. Saying “CFC is not responsible for the greenhouse effect” is **false**.
- Domestic fridge coolants are often sold under names like **Mafron** (halocarbon refrigerants). Large industrial plants may use **ammonia**.
- Classic CFC uses: refrigerators/AC, foam blowing, aerosol pressurising agents, cleaning some electronic components — **not** tubeless-tyre manufacture as a standard CFC use, and **not** automobile exhaust as the primary CFC source.
- Extra UV-B raises risks of **skin cancer**, cataracts, crop damage, and marine plankton stress.

- Depletion is not the same as the greenhouse effect, though some substitutes (HFCs) are strong climate gases.

> **Logic:** Depletion link to **CFCs** is a direct UPPCS fact. Skin cancer is the classic UV-harm key — not liver/brain/oral.

36.3 Ozone Hole

Identity: < 220 DU | Antarctic spring | PSCs + polar vortex | Farman 1985 | Tibet 2005

- The **ozone hole** is a region of severely thinned column ozone, defined as **below 220 DU**.
- It is **not** a literal hole in the sky.
- Classic location: **Antarctic spring** (**September–October**, Southern Hemisphere). Ozone forms slowly over poles, so depletion effects are strongest there.
- Extreme cold forms **Polar Stratospheric Clouds (PSCs)** (ध्रुवीय समतापमंडलीय बादल); nitric-acid chemistry on PSC surfaces activates chlorine from CFCs; the **polar front/vortex** isolates air; chlorine becomes highly active when sunlight returns.
- Discovered in **1985** by **Farman, Gardiner, and Shanklin** (British Antarctic Survey) in *Nature*; total-ozone mapping (TOMS-era) supported monitoring.
- In **2005**, **G.W. Kent Moore** (University of Toronto) and colleagues reported a low-ozone / "ozone halo" feature over the **Tibetan Plateau**.
- Arctic thinning occurs but is usually weaker/less persistent than the Antarctic hole.
- There is **no ozone hole over Uttar Pradesh (उत्तर प्रदेश) / India** as an Antarctic-style seasonal feature.

> **Logic:** Hole = Antarctic spring + DU threshold + PSC/CFC chemistry — not "everywhere CFCs are used."

36.4 Ozone Depleting Substances (ODS)

Identity: CFCs / Halon-1211·1301 / CCl_4 / CH_3Br / HCFCs | ODP scale

- **ODS** are chemicals controlled under the Montreal Protocol because they destroy stratospheric ozone.
- **CFCs:** refrigerants, aerosol propellants, foam blowing, electronics cleaning — long-lived chlorine sources.
- **Halons (e.g. Halon-1211, Halon-1301):** fire suppressants — bromine; often higher ozone-depleting impact per molecule.
- Also controlled: **carbon tetrachloride (CCl_4)**, **methyl chloroform (1,1,1-trichloroethane)**, **methyl bromide (CH_3Br)**, and transitional **HCFCs**.
- **ODP (Ozone Depleting Potential)** ranks destructiveness relative (सापेक्ष) to **CFC-11 = 1**.

- **HFCs** were ozone-friendly replacements but have high **GWP**; controlled later by the **Kigali Amendment** (climate, not classic ODS).
- **Nitrous oxide (N_2O)** also contributes to ozone chemistry / long-term ozone loss discussions, but CFCs/halons dominate classic "ozone hole" stems.

ODS family	Typical use	Fact
CFCs	Cooling, aerosols, foam, electronics cleaners	Main Cl source
Halons (1211, 1301)	Fire fighting	Br; high ODP
CCl_4 / methyl chloroform / CH_3Br	Solvent / fumigant	Controlled ODS
HCFCs	Transitional coolants	Lower ODP; being phased out
HFCs	Coolants	Not ODS; Kigali / GWP

> **Logic:** HFC ≠ ODS, but still relevant under Montreal's Kigali track. Methyl chloroform **is** an ODS under Montreal (ignore any bank key that treats it as non-ODS).

36.5 Montreal Protocol

Identity: Signed 16 Sep 1987 | Force 1 Jan 1989 | Binding ODS phase-out | Ozone Day

- The **Montreal Protocol on Substances that Deplete the Ozone Layer (1987)** is the binding global treaty to phase out ODS (CFCs, halons, CCl_4, etc.).
- It operates under the **Vienna Convention** framework.
- Signed on **16 September 1987**; entered into force on **1 January 1989**.
- **16 September** is observed as **International Day for the Preservation of the Ozone Layer (World Ozone Day)** — UN observance from **1994**.
- Key amendments: London 1990, Copenhagen 1992, Montreal 1997, Beijing 1999, **Kigali 2016 (HFCs; force 1 January 2019)**.
- Often called the most successful environmental treaty — near-universal ratification; ODS production/consumption cut dramatically from peak.
- India has an **Ozone Cell under MoEFCC**; CFC phase-out completed on developing-country schedule; HCFC phase-out continues; Kigali HFC schedule applies later.

PYQ — UPPCS Prelims 2024, Q35

Which agreement is related to climate change (जलवायु परिवर्तन) mitigation?

Options: A. Montreal Protocol B. Ramsar (रामसर) C. Kyoto Protocol D. CITES (साइट्स)

Show answer

Ans: C — Montreal = **ozone**, not climate mitigation.

PYQ — UPPCS Prelims 2019, Q132

Which is NOT correctly matched?

Options: A. Cartagena — Biosafety B. Stockholm — POPs C. Montreal — Ozone layer D. Kyoto — Save water

Show answer

Ans: D — Kyoto = climate/GHG, not “save water.” Montreal–ozone is correct.

PYQ — UPPCS Prelims 2021, Q114 / 2022, Q82 / 2025, Q62

World / International Ozone Day date:

Show answer

16 September (Montreal signing date).

> **Logic:** Montreal = ozone. Kyoto = climate. Never swap. Carbon trading ≠ Montreal.

36.6 Vienna Convention

Identity: 1985 | Framework treaty | Research + cooperation | Parent of Montreal

- The **Vienna Convention for the Protection of the Ozone Layer (1985)** is the **framework** treaty.
- It commits parties to protect the ozone layer through research, monitoring, information exchange, and cooperation.
- It does **not** itself set the detailed binding ODS phase-out schedules — that is Montreal’s job.
- Sequence fact: **Vienna 1985 → Montreal 1987**.
- Together they form the global ozone regime administered with UNEP (संयुक्त राष्ट्र पर्यावरण कार्यक्रम) support (Ozone Secretariat).

Treaty	Year	Role
Vienna Convention	**1985**	Framework / cooperation
Montreal Protocol	**1987**	Binding ODS controls
Kigali Amendment	**2016** (force **1 Jan 2019**)	HFC phase-down under Montreal

> **Logic:** Vienna alone ≠ CFC ban list; Montreal adds the teeth.
